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Chain Rule of Derivatives

⇒ Chain Rule is used when one funⁿ is inside another function.

$$\Rightarrow y = f(g(x))$$

$$\therefore \frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

⇒ Ex:

$$y = (x^2 + 3)^4$$

$$\Rightarrow u' = 2x$$

$$\therefore u = x^2 + 3 \rightarrow \text{inner} \Rightarrow u' = 2x$$

$u' \rightarrow \text{outer}$

$$\therefore 4(x^2 + 3)^3 (2x) = 8x(x^2 + 3)^3$$

Composition of 3 or More Functions

$$\Rightarrow y = \sin((x^2 + 1)^3)$$

$$= \cos((x^2 + 1)^3) \times 3(x^2 + 1)^2 \times (2x)$$

$$= 6x(x^2 + 1)^2 \cos((x^2 + 1)^3)$$

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Chain Rule:

$$\Rightarrow y = (x^3 + 2)^7$$

$$\therefore y' = 7(x^3 + 2)^6 (3x^2)$$

$$= 21x^2(x^3 + 2)^6$$

$$\Rightarrow y = \sin(4x^2)$$

$$\therefore y' = \cos(4x^2) \times 8x$$

$$= 8x \cos(4x^2)$$

$$\Rightarrow y = \ln(2x^2 + 1)$$

$$\therefore y' = \frac{1}{2x^2 + 1} \times 4x$$

$$= \frac{4x}{2x^2 + 1}$$

$$\Rightarrow y = e^{5x^3}$$

$$\therefore y' = e^{5x^3} \times 15x^2 = 15x^2 e^{5x^3}$$

$$\Rightarrow y = \sqrt{3x+2}$$

$$\therefore y' = \frac{1}{2}(3x+2)^{-1/2} (3)$$

$$= \frac{3}{2\sqrt{3x+2}}$$

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